JOHN DEERE

Standard

Process Procedure

DLW25046

Mid Tractor Platform
Manufacturing and Material Requirement for Harness Design and Drawings

Content

		Page
1	Scope and Application	2
2	Definitions and List of Abbreviation	2
3	Verify Compliance with Requirements, Performance, Tests	2
4	DLW-History and Description of Changing	2
5	Requirements	3
6	Other Provisions	18
7	Measures to Verify Compliance with Requirements	18
8	Testrequirement	18

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The originator is responsible for professional content.

The superior of the originator hereby confirms that he has also released the content of this document.
Supervisor: Ulf Hoeppner

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1. Scope and Application

This document describes manufacturing requirements for harnesses, in addition or deviation to existing requirements in JDS-G208 1 to 8. **Variable information must be called out on the drawing.** Further it is also necessary for Harness Drawings that are created by supplier.

2. Definitions and List of Abbreviation

Each drawing must contain a reference to this DLW document.

Manufacturing Requirements according DLW 25046 REV A

Symbol 2.1

3. Verify Compliance with Requirements, Performance, Tests

When using the symbol 2.1, care should be taken to use the current symbol that matches the header of the symbol with the symbol name in the library. ATTENTION: For update of drawings the old symbol should be kept otherwise it will increase tooling costs! The drawing specifies which DLW is valid. If John Deere use data container (drawing created by supplier) to transmit drawing information, then the main datafile is XML file.

4. DLW-History and Description of Changing

Date	Version	Description of Changing's
2022.08.19	DLW 20787 E	Replaced by DLW 25046
2020.11.20	DLW 20803	Replaced by DLW 25046
2020.05.10	DLW 22788 A	First Edition, Version A

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5. Requirements

5.1 General requirements

Temperature see according drawing spec, specification in degree Celsius. By long term heat ageing of 3000 h.

a. ☒ If not especially called out at single part, all parts must be resistant from -40°C up to 85°C.

b. ☒ If not especially called out at single part, all parts must be resistant from -40°C up to 100°C.

c. ☒ If not especially called out at single part, all parts must be resistant from -40°C up to 125°C.

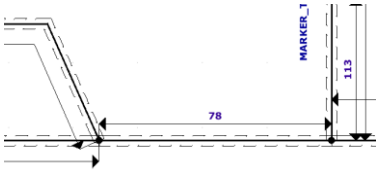
Needed temperature range shall be called out on the drawing.
If not already specified by part numbers, the applied temperature class shall be appropriate.

5.2 CPA

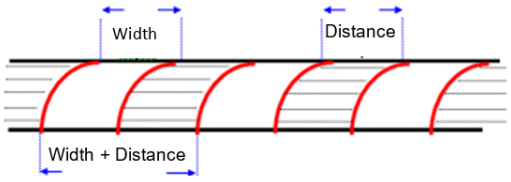
CPA (Connector position assurance) and locking lever shall be open in delivery condition. If closed position is desired, it shall be explicitly stated on the drawing.

5.3 Spiral Tape

Textile tape in black, width of tape according manufacturers choice, distance is min. width of tape and no bigger than 2 times from chosen tape width.
Exception: If bundle diameter too small for spiral tape (up to 3 wires), use full tape.
At beginning and end of spiral tape wrapped twice.



Symbol-5.3.1



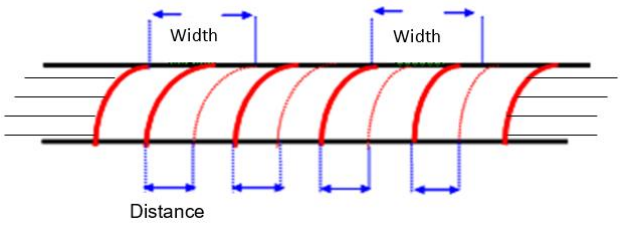
Symbol-5.3.2

5.4 Full Tape (overlap tape)

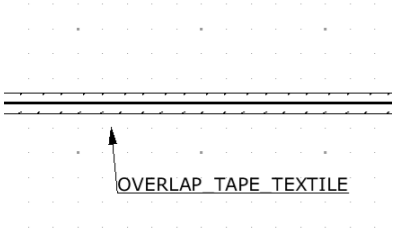
Textile tape in black, width of tape according manufacturers choice, distance of tape is minimum 20% and maximum 50% of tape width.

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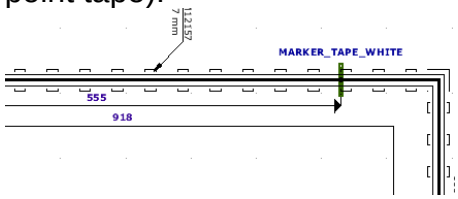
Symbol-5.4.1



Symbol-5.4.2

5.5 Space Tape

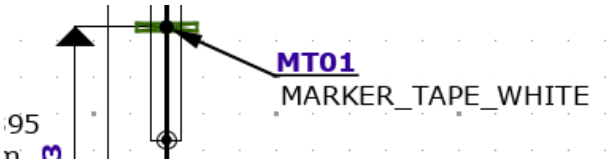
Textile tape in black, width of tape according manufacturers choice, wrapped min two times every 150 mm (max.), minimum 2 space tape on a bundle (up to 120 mm length is one spot tape enough), first and last place of space tape is 30 mm after and before a break-out point. (Space Tape is same like a recurrent spot tape or recurrent point tape).



Symbol-5.5.1

5.6 Marker Tape ⊠

PVC tape in the specific color according to the drawing requirements (data container), width 15 - 25 mm (9mm width of tape is also allowed, when space is not enough), wrapped min. two times.



Symbol-5.6.1

If color of marker tape is not specified on drawing, use white tape. With library name or spool. For example MT##

5.7 Spot Tape ST##

Textile tape in black according to the drawing requirements (data container), width 15 - 25 mm, wrapped min. two times. Exception: Up to 3 wires in bundle, it is also allowed to use textile tape with 9mm width.

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5.8 Label Tape (Assembly)

The label has to be fixed on the harness as a flag, if not otherwise specified. All imprinted labels have to be readable from both sides of flag. Width of flag shall be between 10 and 30 mm, font arial size 8. If QR code is printed, it has to have the size of min. 10 x 10 mm.



Symbol-5.8.1



Symbol-5.8.2

5.9 Barcode Label

The Barcode Label has to be fixed on the harness as a flag with following information:

- John Deere part number with revision level
- DL numbers
- Manufacturer name identification
- Manufacturing date: Month Day Year DDMMMYYYY (24Jan2019)
- The John Deere harness part number also imprinted (bigger than other informations) on the label as additional barcode. Coding has to be located next to the part number.
- The area of label on harness has to be free or without John Deere required information. See below
- Size of label has to be between 80-120 x 40-60 mm, font arial size 8. Size of Barcode min. 30 x 10 mm, QR code 10 x 10 mm
- Flag has to be large enough to show the complete barcode on it

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Company Use

5.9.1 Barcode label as a flag



Symbol-5.9.1.1

Example:



Symbol-5.9.1.2



Symbol-5.9.1.3



Symbol-5.9.1.4

5.9.2 Barcode label as sticker cable channel



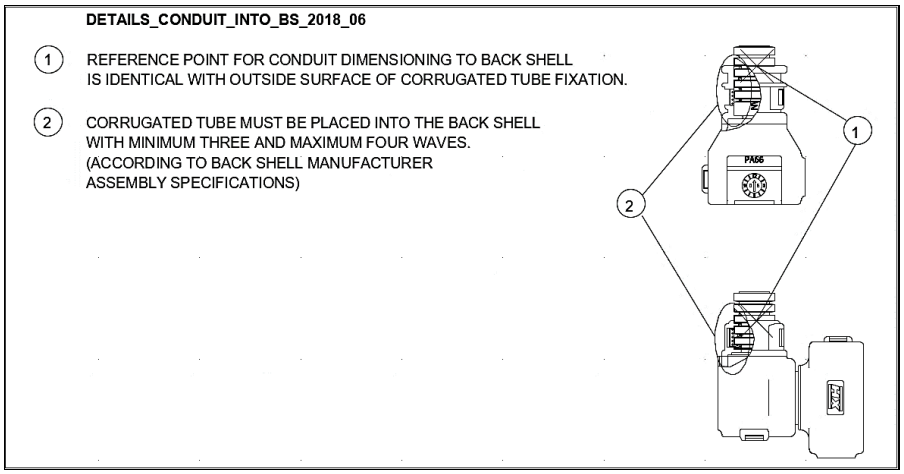
Symbol-5.9.2.1

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5.10 Plastic backshells

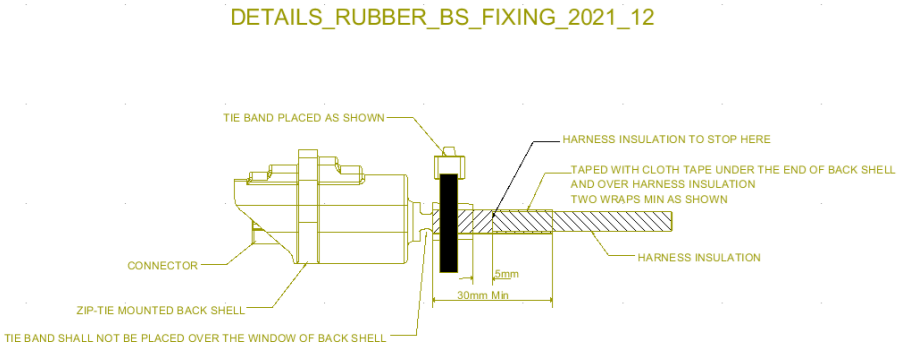
Backshell has to fit to conduit and connector. If no other specification of backshell supplier is available, holding rip of backshell has to be for recesses min. of three max. four.



Symbol-5.10.1

5.11 Rubber backshells

According to part number on the drawing, attached to the corresponding component according to drawing requirements (data container).
Specification see below



Symbol-5.11.1

5.12 Grommet

In addition to part number and position on the drawing (data container) the orientation of the grommet shall be taken from 3D file (pvz). Secure against slipping, except not fixed is call out on drawing. If tightness is required so it has to be called out on drawing. If own clamping force is not sufficient use tape under it or secure it with tie band (See clause 24). Reference point for dimensioning according DLW25043. In areas where the grommet doesn´t need to be fixed, it has to be called on drawing out as a note.

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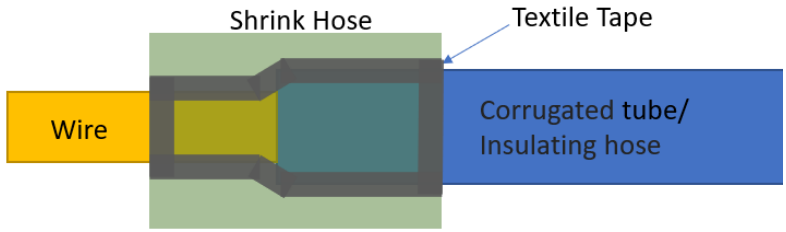
Company Use

5.13 Dimension in brackets

Dimensions in brackets are informational only. They are not necessary for the quality.

5.14 Heat-shrink Tube (on ground cables)

Diameter of heat-shrink tube as small as possible, adopted to bundle diameter.
If necessary, use tape under shrink tube to secure against slipping. If SH with glue is specified, the dimension of 40 mm shall be maintained. If necessary, tape can also be used underneath the heat shrink hose. If not specified on the drawing than:
Length of shrink tube 40 mm, 20 mm on each site.
Should match to the temperature of the cable set. Post-shrinking under installation conditions (in the temperature range of use) shall not rip the shrink tube.



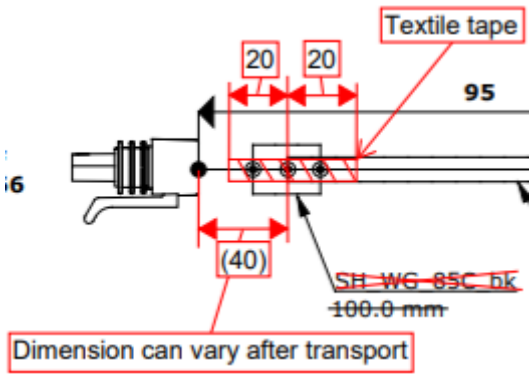
Symbol-5.14.1

5.15 Conduit Break-out Points

Spot tape shall be placed over wires at the end and at the beginning of the conduit. They shall be fixed with heat shrink tube on the branch in accordance with the drawing requirements (data container).

5.15.1 Conduit (Corrugated tube) not fixed with heat shrink tube or tape.

Spot tape shall be placed on the cables at the end of the conduit, min. 20 mm to each side, wrapped three times to secure the wires insulation against scoring. Dimension can vary after transport if it is called out in a clamp.



Symbol-5.15.1.1

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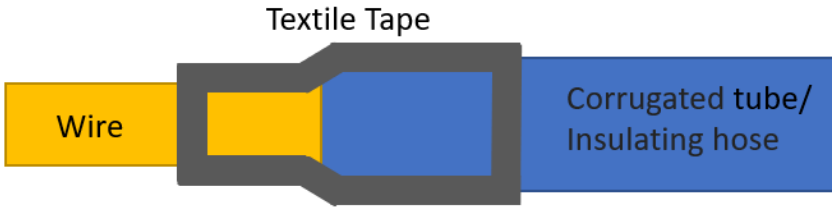
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Symbol-5.15.1.2

5.15.2 Conduit (Corrugated tube; Insulating hose) fixed with tape.

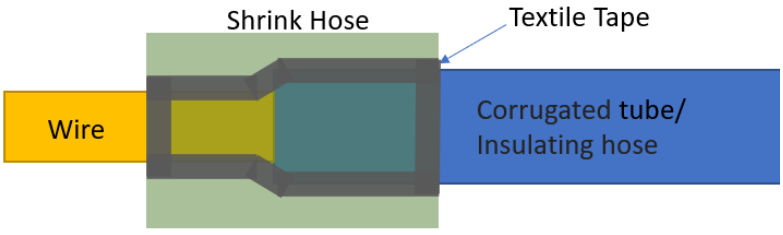
Spot tape shall be placed on the fixed tube and cables. From end of conduit min. 20 mm to each side, wrapped three times.



Symbol-5.15.2.1

5.15.3 Heat-shrink Tube

Diameter of heat-shrink tube as small as possible, adopted to bundle diameter. If necessary, use tape under shrink tube to secure against slipping.
If SH with glue is specified, the dimension of 40 mm must be maintained. If necessary, tape can also be used under the shrink hose. If not specified on the drawing than: Length of shrink tube 40 mm, 20 mm on each side. Should match to the temperature of the cable set. Post-shrinking under installation conditions (in the temperature range of use) must not rip the shrink tube.



Symbol-5.16.3.1

5.16 Fixed Tube

5.16.1 Slit Conduit (Corrugated tube)

In addition to the conduit ends which are covered with points above, spiral or space tape on slit corrugated tube. If required, further definition will be given. Otherwise use the supplier procedure.

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5.16.2 Unslit Conduit (Corrugated tube)

Fixed with heat shrink tube on the branch in accordance with the drawing requirements (data container), plus spot tape over wires at the end and the beginning of the conduit, min. 20 mm both sides of the tube opening.

5.16.3 Double Tube Conduit (Corrugated tube)

In addition to the conduit ends which are covered with points above
Fixed with heat shrink tube on the branch in accordance with the drawing requirements (data container), plus spiral taping under conduit tube, plus spot tape over wires at the end and the beginning of the conduit, min. 20 mm both sides of the tube opening.



Symbol-5.17.3.1



Symbol-5.17.3.2

5.17 Meshed Tube (Braided sleeve or hose)

Fixed with tape on the branch in accordance with the drawing requirements (data container). If HSH is needed use HSH with glue or tape on wires and tube.



Symbol-4.18.1

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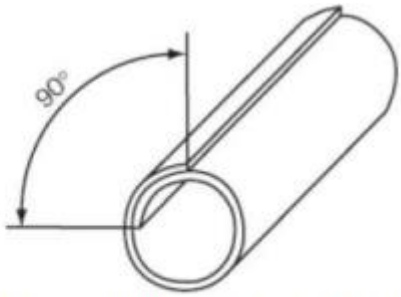
Company Use

5.18 Self-wrapping sleeve (old DLW 20803)

5.18.1 Each drawing which contains the self-wrapping sleeve shall include a reference to this DLW document.



5.18.2 Symbol-5.19.1 The self-wrapping sleeve construction profile shall be as shown in Figure 1.



Skizze 1: selbstschließende Gewebeschutzhülle Konstruktionsprofil
Figure 1: Self-wrapping sleeve construction profile

5.18.3 The size of the self-wrapping sleeve shall be chosen in a kind, that after the covering of the bundles the overlapping of the sleeve is min 25%. On drawing is a generic Number given. Suppliers have to choose the correct diameter of self-wrapping sleeve.

5.18.4 The self-wrapping sleeve shall be twisted to achieve the fill rate of 95%-100%. The sleeve has to be twisted three to four times per meter. Nevertheless, the number of twists can be increased to adapt the diameter of sleeve to a smaller bundle.

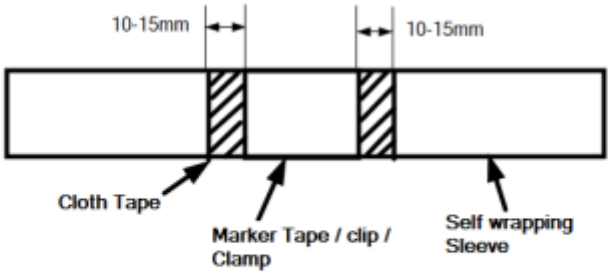
5.18.5 Wires shall not be visible at any point through the seam of the self-wrapping sleeve. This requirement applies to the assembled and the unassembled wire harness. Set every 150 mm a spot tape, then it is ensured.

5.18.6 Over self-wrapping sleeve, two wraps of textile tape shall be present under clamps, clips, tie bands extending out to 10-15mm on each side. This will prevent wear-marks from appearing on the insulation of the wire. It is allowed to add spot tapes beneath Self-wrapping sleeve to prevent wires protruding from the sleeve.

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5.18.7 Over self-wrapping sleeve, two wraps of textile tape shall be present under marker tape location extending out to 10-15mm on each side. See Figure 2.

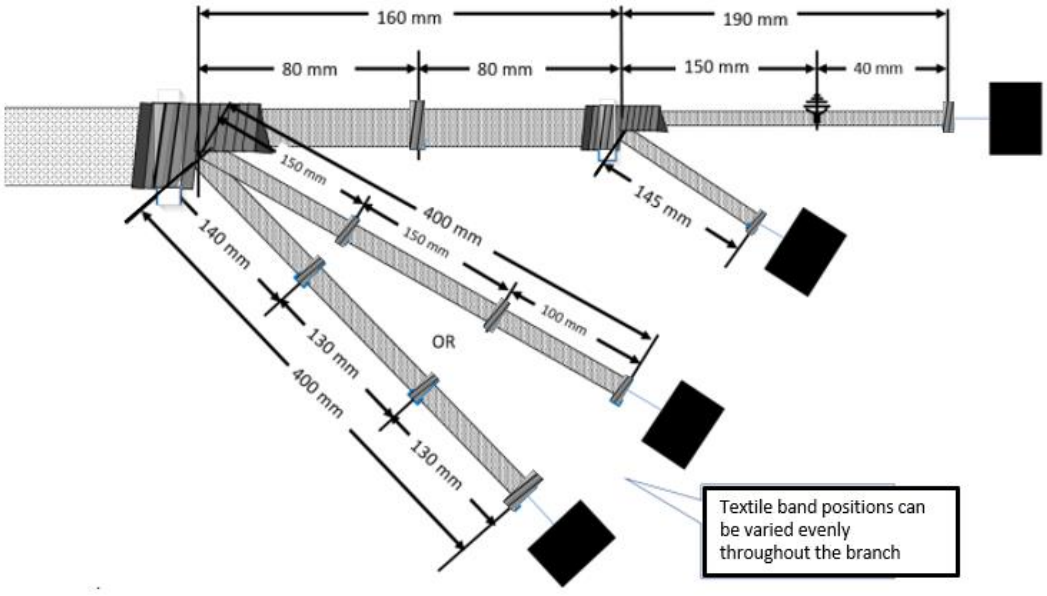


Slizze 2: Kleiderband über Selbstschließende Gewebeschutzhülle
Figure 2: Cloth tape over Self Wrapping Sleeve

5.18.8 Tag end of tape shall wrap on itself and not on the self-wrapping sleeve.

5.18.9 At the end of a branch self-wrapping sleeve shall be wrapped with two layers of textile tape.

5.18.10 Unless otherwise specified on drawing, the self-wrapping sleeve shall be taped (spot tape) at 150 mm intervals. It is also allowed to use spiral tape.

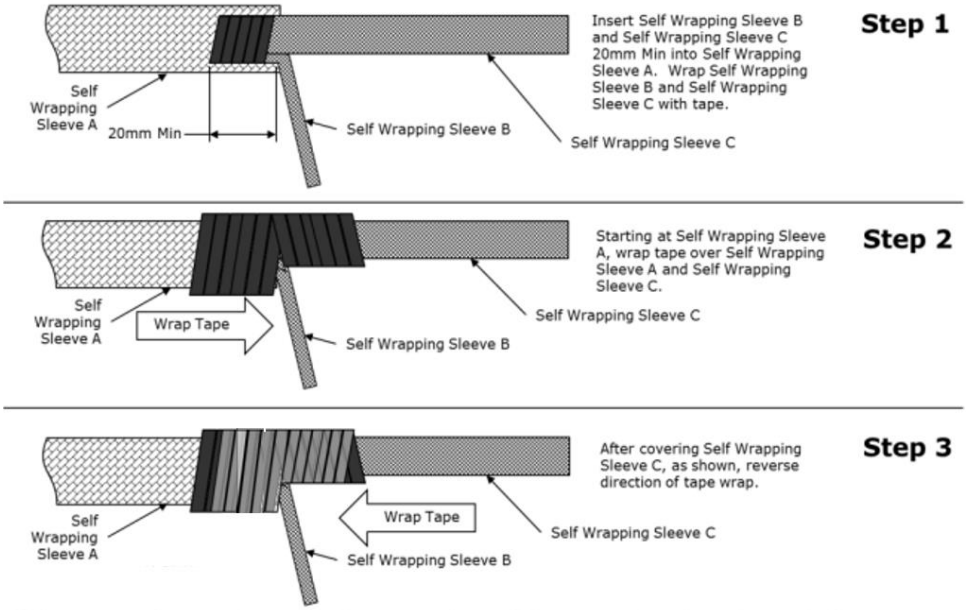


Skizze 3: Selbstschließende Gewebeschutzhülle
Figure 3: Self wrapping sleeve

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5.18.11 Self-wrapping sleeve with a parallel break-out

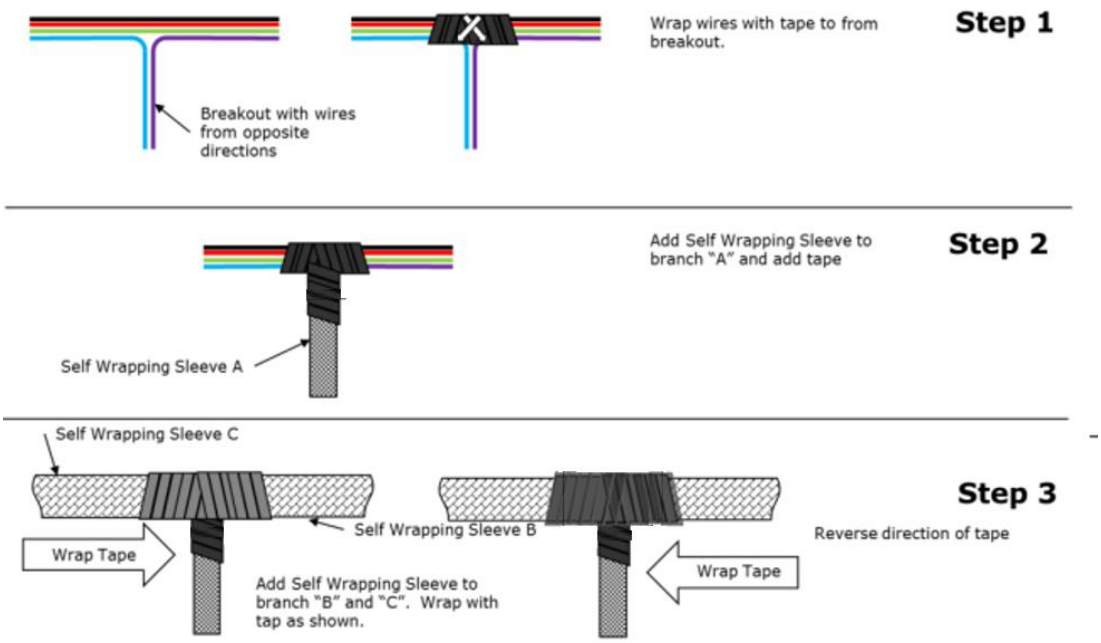


Skizze 4: selbstschließende Gewebeschutzhülle – parallele Kabelausbindung
Figure 4: Self-wrapping sleeve – parallel wire break-out

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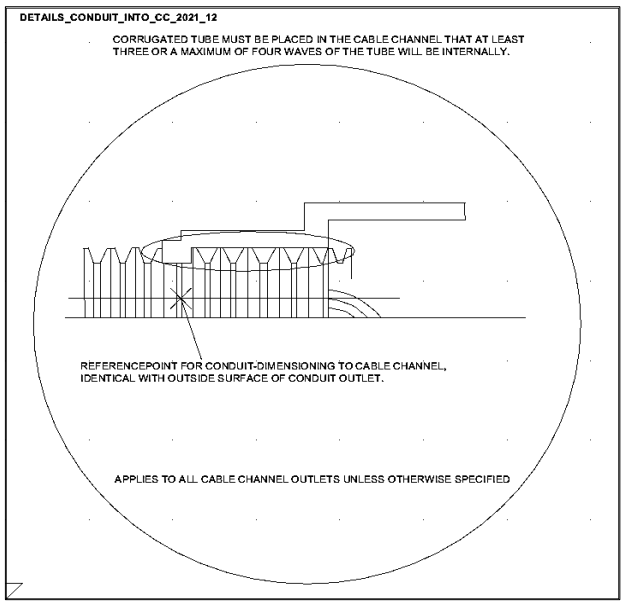
5.18.12 Self-wrapping sleeve with rectangular wire break-out



Skizze 5: selbstschließende Gewebeschutzhülle – rechtwinklige Kabelausbindung
Figure 5: Self-wrapping sleeve – rectangular wire break-out

5.19 Cable Channel, T- and Y- Pieces

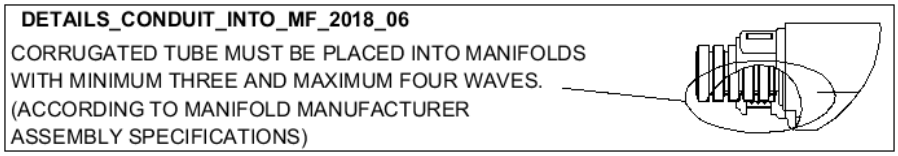
For clear dimension reference points and further Conduit brake out points see DLW25043 “Dimensional references for harness connectors”.



Symbol-5.20.1

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Symbol-5.20.2

5.20 Cable Tie

In accordance with part number on the drawing including orientation (of the cable tie) according to drawing or 3D-pvz file, always fixed/cut by a cable tie pistol, plus spot tape over wires.

on wires

Spot tape over wires to protect the insulation. No permanent deformation

on conduits, connectors and rigid parts (no slit conduit, no PVC tube)

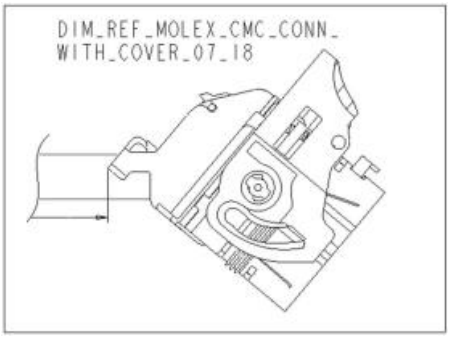
No permanent deformation

$$\text{recommended pulling force} = \frac{\text{minimum holding force}}{2}$$

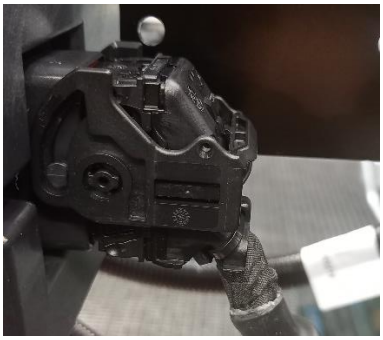
Symbol-5.21.1

5.21 Wire branch retention on connector cover

Tape wire bundle to secure wire insulations, fix with cable tie, if not other specified.



Symbol-5.21.1



Symbol-5.21.2



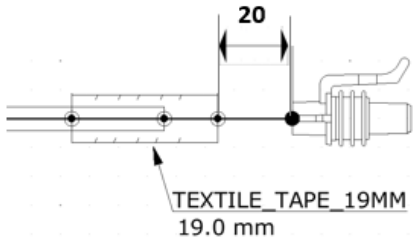
Symbol-5.21.3

5.22 Free Wire length in front of connector

The free length of the wires between connector housing and the closest harness coating is 20 mm. Exception: if more than 8 wires given, then it is allowed to take a free wire distance of 40 mm to connector. The overall internode bundle dimension shall stay same.

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Symbol-5.22.1

5.23 Manifold

Harness shall be wrapped with tape (at least spiral tape, textile tape) Within manifolds to avoid squeezing of tree wires while manifold closing.

5.24 Fixation

If it is necessary use textile tape under heat shrink hose without glue, grommet outlets and underneath fixation elements to ensure it in place even after transport.

5.25 Splices

Splices shall be implemented using JDS-G208.1 and JDS-G208.3. All splices have to be insulated with a self-adhesive shrinking hose. Contrary to JDS-G208.1 and JDS-G208.3 all crimped splice connections shall not be soldered. For can bus splice DLW25025 has to be considered.

5.26 Ringterminals

Wire and insulation crimp for ringterminals shall be covered/insulated with a shrinking hose. Min. extended length beyond insulating crimp is the cable diameter plus 5mm. Contact area shall be free from insulation according to DIN 46225.

5.27 Connectors used for Automotive Ethernet

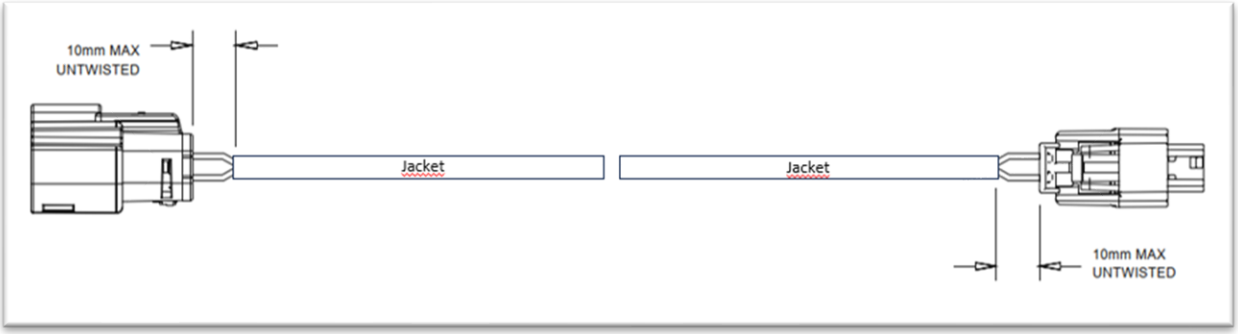
Wire types that are used for ethernet applications can be identified from the multicore table by the description ETH or ETHERNET.

Connector systems dedicated for automotive ethernet applications shall be processed according to the application specification of the connector manufacturer. Example: Application of connector 57M16487 (TYCO: 1-2292937-1) according to TYCO specification 114-151045.

If the connector system is not dedicated for automotive ethernet application following guideline shall be fulfilled:

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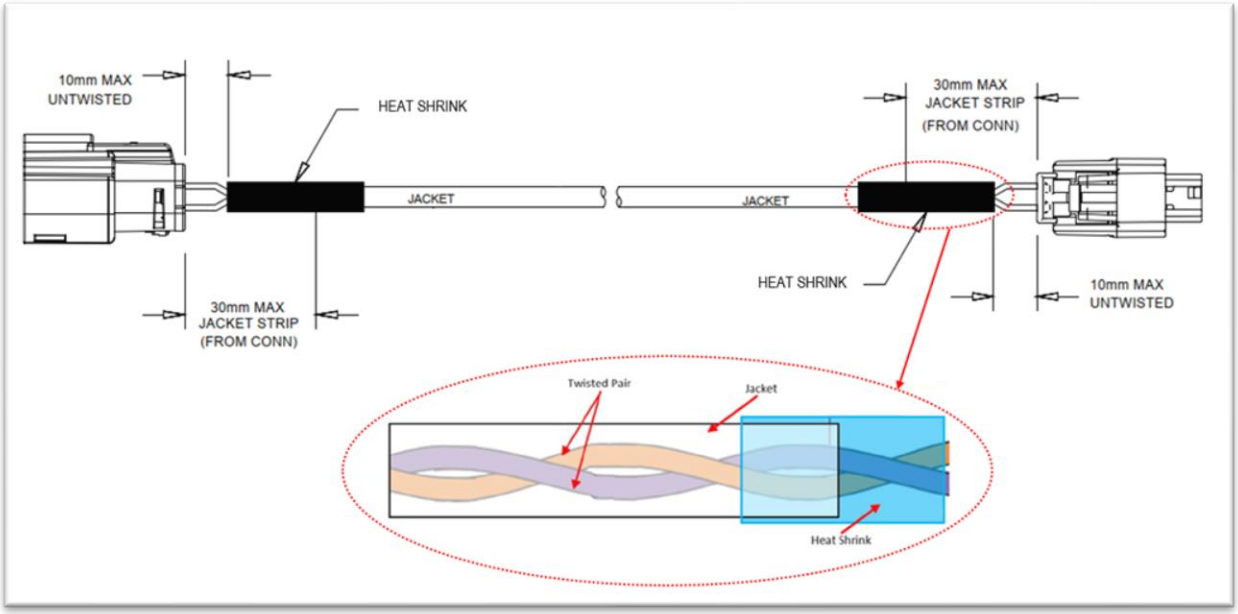
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Symbol-5.27.1

The untwisted wires length of the ethernet cable measured from the back of the connector to the cable jacket shall not exceed 10mm.

If this requirement cannot be met, the deviation described below shall be followed after it has been discussed and agreed with John Deere.



Symbol-5.27.2

If the untwisted wires length is not within 10mm, the jacket can be stripped up to 30mm from the back of the connector. The wires, entering device or inline connectors, shall be manually retwisted to within 10mm or less from the back of the connector and held with a glue-filled heat shrink.

Bending radius tighter than four times cable diameter are not permitted and the cable must not be looped in any harness bundles to take up excess length.

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6. Other Provisions

This document describes requirements for harnesses, that apply or deviate from existing requirements from JDS-G208.1 to 8

1. Crimp connections implemented according to IEC60352-2.
2. Wires according to DIN 72 551 AND ISO 6722.
3. Delivery
   General packaging according to JDV32.
   Harness packaging as per JDV32.1/3
4. Twisted cable for CAN bus implemented using ISO 11783-2
5. Also note the DLW25043
6. Drawing has to be seen in conjunction with 3D model. All information about component orientation and additional assembly information is given in 3D model.
7. Rounding of dimensions last place
   8 – 2 to 0
   3 – 7 to 5
8. Four digit circuit no. and suffix must be marked consecutively, beginning at one side, with max 25mm from connector back and max. 40 mm intervals in between.
9. Contrary to JDS.G208-3, circuit color print white on wires is only needed for brown, black or two color wires, where basic color is black or brown.
10. Materials per JDM H31

7. Measures to Verify Compliance with Requirements

Within supplier quality audits, DPAR and PPAP review as well as incoming goods inspection, that are described within the internal standard JDS-G223 X3 and JDS-G223.

8. Testrequirement

Electrical test implemented as per JDS-G208.5., circuit continuity testing additionally with LAN962, PART 32. Specialized test adapters shall be used for electrical test to avoid damage to the pins and terminals (e.g. sizing, bending, scratching of surface, etc.). The male pin shall not be inserted into the female test terminal, as well as no male test pin shall be inserted into the female terminal. In minimum Spring Loaded Pins shall be used.

9. Documentation and History

Reference to the points above

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